Tensor, Low-Rank, and Sparse Inference Systems

Literature synthesis, completed design work, and proposed research portfolio

Research briefing for advisor and collaborators
Evidence cutoff: 23 July 2026

Current recommendation. The field has moved beyond simple KV cache compression. Strong work now co-designs the representation, RoPE treatment, rank allocation, residual protection, and the attention kernel. Our recommended first implementation is **SAGE-CP-KV**: a Softmax-Aware CP tensor representation for the compressible bulk, plus attention-influence-selected RoPE-pair sparse corrections, consumed directly without reconstructing old keys or values.

Novelty warning. "Low rank + sparsity" is not itself a novel claim. GEAR already uses low-rank and sparse corrections to quantization error; OjaKV keeps exact anchors around a low-rank cache; SALS and ShadowKV use low-rank structure for sparse access; JoLT pairs Tucker factors with a low-bit residual. A publishable paper must make the *objective, sparse structure, tensor execution, or runtime policy* genuinely different and demonstrate a new quality–memory–latency Pareto point.

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1 Scope, method, and how to read the evidence

This report surveys methods whose main mechanism compresses the *feature/channel structure* of keys or values using a low-dimensional matrix, tensor, dictionary, or latent representation. It also includes the closest hybrids—quantization, sparse selection, exact anchors, architecture conversion, and residual coding—when they define the novelty boundary for low-rank work. Pure token-eviction and pure quantization papers are included only as required baselines or as evidence about failure modes. This is therefore a comprehensive *scoped* survey, not a claim to enumerate every paper that contains the words "KV cache."

Evidence-status legend

P = peer-reviewed archival conference/journal or accepted poster; **A** = arXiv or public submission/preprint; **U** = withdrawn or desk-rejected public prior art. Preprints are useful for novelty checking, but their performance claims require independent reproduction. The survey separates reported results from this report's inferences.

Sources were checked against primary records: ACL Anthology, PMLR, NeurIPS proceedings, OpenReview, and arXiv. The cutoff matters because several directly relevant results appeared in 2026, including OjaKV, StiefAttention, KV-CoRE, STAR-KV, RAP, and JoLT. Numerical claims below are the authors' reported results and are not directly comparable unless model, workload, context, precision, hardware, and memory accounting match.

2 Current state of the project

Work completed to date

1. We completed a primary-source literature synthesis of low-rank, latent, tensor, sparse-residual, and closely related KV-cache methods through the stated cutoff. The resulting taxonomy and 27-method landscape explicitly separate peer-reviewed evidence from preprints and public unpublished prior art.
2. We derived the native KV memory model, separated weight-side "born compressed" methods from activation-side runtime fitting, and identified RoPE handling, dense reconstruction, and fit-cost amortization as the three main feasibility constraints.
3. We established the novelty boundary: low rank, tensor decomposition, sparsity, an exact window, or compressed-domain attention is not individually new. The candidate contribution must come from their specific mathematical and systems co-design.
4. We developed four complementary research concepts: the single-device **SAGE-CP-KV** method and the collaborative-inference extensions **LoSpar-CI**, **Shufflerless-CI**, and **SparsePipe**. Separate detailed proposal drafts also exist for LoSpar-CI, Shufflerless-CI, SparsePipe, CATT-KV, and CP-Direct-KV.
5. For SAGE-CP-KV we derived a CP tensor representation, an attention-influence sparse selection objective, reconstruction-free key-score and value-aggregation equations, a physical memory model, a payback admission rule, theoretical targets, and falsification experiments.
6. We defined staged experimental gates and same-byte baselines so that a negative result stops systems implementation early rather than being hidden by nominal compression ratios.

Work not yet completed—and therefore not claimed as a result

We have not yet produced CP/TT rank measurements on recorded model caches, trained projection factors, implemented fused Triton/CUDA kernels, proved the proposed error bounds, integrated the methods into an inference runtime, or measured end-to-end speed, memory, energy, or downstream quality. All numerical thresholds in the proposal sections are *decision gates or targets*, not achieved results. This distinction should be preserved in presentations and manuscripts.

2.1 The main four research ideas

Idea	Primary setting	Core scientific question	Recommended role
SAGE-CP-KV	One accelerator; long-context decoding	Can a CP bulk plus structured sparse innovations improve the quality–memory–latency frontier without dense reconstruction?	**First implementation and first paper candidate**
LoSpar-CI	High-dimensional tensors crossing device boundaries	Can communication scale with intrinsic tensor rank and sparse innovation count, with certified progressive refinement?	Separate cross-domain method after structural pilot
Shufflerless-CI	Heterogeneous edge LLM pipeline	Can representation and layer placement jointly guarantee stationary KV residency and remove CPU–GPU/device shuffling?	Systems extension after the single-device kernel works
SparsePipe	Dynamic heterogeneous pipeline	Can sparsity level, wire format, and placement be scheduled together under live load and an accuracy constraint?	Broader, higher-risk follow-on systems project

3 Why the KV cache hits the memory wall

For batch size B, layers L, KV heads H_{kv}, sequence length S, head dimension d, and b bytes per element, the native cache occupies

$$M_{\mathrm{KV}} = 2BLH_{\mathrm{kv}}Sd\,b,$$

where the factor two represents keys and values. During each decode step, attention reads old cache entries again. Thus long-context inference is limited by both **capacity** and **HBM bandwidth**; reducing bytes stored is not enough if decompression adds more work or materializes a dense cache.

For one head, $K, V \in \mathbb{R}^{S\times d}$ and query $q \in \mathbb{R}^d$,

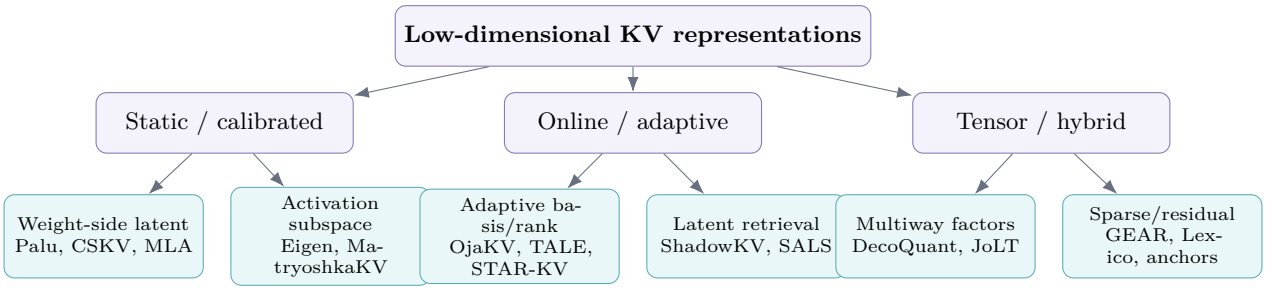
$$\mathrm{Attn}(q, K, V) = \mathrm{softmax}(qK^\top/\sqrt{d})V.$$

A matrix approximation stores $K \approx AB^\top$, with $A \in \mathbb{R}^{S\times r}$, $B \in \mathbb{R}^{d\times r}$. It is useful only if $r(S + d) < Sd$ after metadata and precision are counted. More importantly, a system win requires computing $qBA^\top$ and the value aggregation without writing an $S \times d$ reconstruction. The same distinction governs tensor methods.

Four independent design axes

1. **What is factorized?** Projection weights, realized activations, attention scores, or a multiway cache tensor.
2. **When?** Training time, post-training calibration, prefill, once per aged block, or periodically during decode.
3. **What remains exact?** Recent tokens, anchors, outlier entries, RoPE dimensions, or selected tokens.
4. **How is it consumed?** Dense reconstruction, reconstruction of a selected subset, or direct compressed-domain attention.

4 Taxonomy of low-rank KV compression



4.1 Static activation-space projection

Eigen Attention learns/calibrates low-rank subspaces and evaluates attention in the reduced space. It established that feature reduction can lower both cache size and attention latency, reporting up to 40% cache reduction

and 60% attention-latency reduction [10]. Its limitation is a static subspace and a reconstruction-oriented proxy rather than an objective tied to the final decoder output.

MatryoshkaKV learns orthogonal projections with nested usable ranks, allowing ranks to vary across heads and layers without training a separate basis for each budget [17]. This makes allocation flexible, but still relies on trained static feature spaces. **StiefAttention** directly attacks the objective mismatch by optimizing orthonormal bases on a Stiefel manifold to minimize decoder-layer output error, then builds layerwise error–rank profiles [28]. It is a strong 2026 preprint baseline for any "attention-aware" claim. **Swift-SVD**, accepted at ICML 2026, gives an activation-aware closed-form spectral solution, incremental covariance aggregation, and dynamic rank allocation for joint weight/KV reduction [29]; it is a particularly strong calibration-time matrix baseline.

KQ-SVD argues that approximating keys alone is not optimal for preserving the attention matrix and gives a closed-form decomposition targeting query–key interactions [24]. **CurDKV**, although it retains selected rows rather than storing factors, uses CUR leverage scores and value information to preserve attention output; it shows why attention scores alone are an insufficient importance target [23]. Together these works shift the standard from raw Frobenius error toward attention or decoder-output fidelity.

4.2 Weight-side and "born compressed" representations

Palu factorizes key/value projection weights so low-dimensional latent states are cached; its kernel work demonstrates the importance of fusing projections rather than reconstructing the cache [16]. **LoRC** similarly performs post-training weight-level factorization, but allocates layerwise sensitivity progressively because errors accumulate through depth [14]. **CSKV** uses low-rank K/V layers, a small exact recent branch, and lightweight reconstruction training; it reports 80% cache reduction, and up to 95% when combined with quantization [12].

DeepSeek-V2's Multi-head Latent Attention (MLA) is architectural rather than a retrofit: the model is trained to cache a compact latent vector and separates positional information where needed; DeepSeek-V2 reports 93.3% KV-cache reduction relative to its stated comparison [15]. **MHA2MLA** retrofits existing MHA models using joint SVD and partial RoPE with a small amount of adaptation data, reporting 92.19% reduction on Llama-2-7B with a 0.5% LongBench drop [25]. These methods make a central feasibility point: the cheapest runtime decomposition is often the one moved into fixed weights, but RoPE must be explicitly separated, aligned, or otherwise handled.

RAP takes that lesson further: it prunes RoPE-aligned column pairs so transformations can be absorbed and reconstruction eliminated [31]. It is pruning rather than a data-adaptive factorization, but it closes off vague novelty claims based only on "RoPE-aware low rank." **EliteKV** selects head-specific RoPE frequencies and applies joint low-rank K/V projection with limited uptraining, reporting 75% cache reduction in its tested model families [30]. **ReCalKV** differentiates K and V, reorders similar key heads for grouped SVD, and fuses the value-side calibrated matrices to avoid added compute [26].

4.3 Online and token-adaptive low rank

TALE makes ranks token-adaptive, compresses lazily, avoids key reconstruction by treating K and V asymmetrically, and combines the representation with multilevel quantization. It reports 9.1× memory compression and up to 2× latency reduction in its long-context setting [18]. **OjaKV** keeps crucial tokens as exact anchors and adapts the low-rank basis via Oja's online PCA, with a larger prefill update and lightweight periodic decode updates [27]. **STAR-KV** is a 2026 preprint using differentiable thresholding for adaptive head/block ranks, hybrid K/V factorization, mixed precision, and Triton kernels [32]. **VarRate**, a very recent preprint, argues for variable-rate codes rather than uniform rank or irreversible eviction [37].

The lesson is not that online fitting is always best. It is that a static basis can fail under distribution shift, while per-token or per-block adaptation can cost more than it saves. Every online proposal therefore needs an explicit **amortization horizon**: how many future reads remain after a block is compressed?

4.4 Low rank as an index for sparse retrieval

ShadowKV keeps a low-rank key representation on GPU, offloads values, and reconstructs only selected sparse KV pairs; it reports up to 6× larger batch sizes and 3.04× throughput on its A100 evaluations [19]. **SALS** performs RoPE-free sparse token selection in a low-rank latent space and reconstructs only the chosen tokens, reporting 6.4× cache compression and substantial operator/end-to-end gains in its settings [20]. **RocketKV** combines coarse eviction with top-k sparse attention and low-dimensional selection [21]. These systems use low rank primarily to reduce the cost of *finding* important tokens, not necessarily as the final representation from which all attention is computed.

4.5 Tensor decompositions and structured residuals

DecoQuant uses matrix-product-operator/tensor decomposition to isolate outlier-bearing small tensors and make the large tensors easier to quantize; the decomposition assists data-free low-bit compression rather than defining direct tensor attention [13]. **JoLT** is the closest tensor prior: a partial Tucker decomposition compresses token and feature modes while leaving head/layer axes intact, and a Johnson–Lindenstrauss-rotated low-bit residual recovers discarded energy. It reports near-lossless 2–3× compression and a randomized variant that reduces compression time [35]. **Tucker-KV** publicly proposed HOSVD/HOOI refinement and budget allocation, but its ICLR 2026 submission was desk-rejected; it is prior art, not peer-reviewed evidence [36].

GEAR is the decisive prior for low-rank+sparse claims. It first quantizes most cache entries, then approximates quantization error with a low-rank matrix and corrects individual outliers with a sparse matrix [11]. **Lexico** represents each KV vector by sparse coefficients over a universal dictionary, retaining 90–95% of original GSM8K performance using 15–25% of native cache memory in reported experiments [22]. These works show that residual structure matters, but leave open whether a *tensor* bulk plus *attention-influence-structured* sparse residual can be executed directly.

4.6 Diagnostics, negative evidence, and evaluation

KV-CoRE benchmarks data-dependent rank compressibility across layers/domains and relates normalized effective rank to downstream degradation [33]. **Quantization Dominates Rank Reduction**, an unreviewed 2026 preprint, reports that at matched storage low-bit quantization outperforms rank reduction across its tested models; its explanation is that projection can change attention routing while bounded quantization often preserves dimension-wise information [34]. Even if its universality is disputed, it sets the correct baseline challenge: tensor/low-rank methods must beat INT4 at matched *physical bytes*, not just beat SVD.

The Risk of KV Cache Compression studies intrinsic compressibility and worst-case risk [38]. **The Pitfalls of KV Cache Compression** finds uneven instruction degradation and system-prompt leakage under several eviction methods [39]; it motivates protected instructions and multi-instruction evaluation for all compression families. A workload-aware 2026 benchmark finds that compression ratio alone poorly predicts quality or end-to-end performance [40]. The TMLR survey on KV-cache management likewise finds that method choice depends on workload and system constraints [9].

5 Landscape table

Method	Year	Main representation	Adaptation / exact part	Execution idea	What remains open
Eigen Attention	24 **P**	calibrated low-rank subspace	static, layer/head subspaces	attention in reduced space	distribution shift; output-aware objective
GEAR	24 **P**	quantized bulk + low-rank error + sparse outliers	online cache coding	hybrid reconstruction/compute	attention-aware structured sparsity
CSKV	24 **P**	low-rank K/V layers	small full-precision window	low-dimensional cache; light training	direct tensor representation
DecoQuant	24 **P**	MPO-assisted quantization	data-free transform	quantized tensors	runtime factor-domain attention
LoRC	24 **A**	weight low rank	progressive layer allocation	plug-in projection	data-adaptive residual
DeepSeek MLA	24 **A**	trained latent KV	decoupled positional branch	born compressed, absorbed projections	retrofit without broad tuning
MHA2MLA	25 **P**	joint K/V low rank + partial RoPE	data-efficient retrofit	converts MHA/GQA toward MLA	adaptation data and deployment kernels
EliteKV	25 **A**	RoPE frequency selection + joint low rank	limited uptraining	partially restores key linearity	model-family transfer and kernel cost
Palu	25 **P**	factorized K/V weights	calibrated grouping	fused latent computation	adaptive per-input correction
MatryoshkaKV	25 **P**	nested orthogonal projections	trainable rank choices	reduced feature cache	runtime distribution shift
TALE	25 **P**	token-adaptive low rank + quantization	lazy; K/V asymmetric	reconstruction elimination	multiway/tensor residuals
ShadowKV	25 **P**	low-rank keys + offloaded values	sparse selected tokens exact	selective reconstruction	single-device no-offload direct path

Method	Year	Main representation	Adaptation / exact part	Execution idea	What remains open
SALS	25 P	latent low rank for sparse selection	reconstruct selected tokens	RoPE-free latent retrieval	all-token factor execution
RocketKV	25 P	low-dim sparse retrieval	two-stage selection	top-k attention	representation fidelity of unselected tail
Lexico	25 P	sparse universal dictionary	per-token OMP codes	sparse reconstruction	cost of coding; low-rank bulk synergy
CurDKV	25 P	CUR row selection	value-guided leverage	keeps selected KV exact	direct low-rank/tensor storage
KQ-SVD	25 A	attention-matrix low rank	query calibration	closed-form target	runtime and tensor extension
OjaKV	26 P	online PCA basis + anchors	prefill + periodic Oja updates	FlashAttention compatible	update payback; tensor factors
StiefAttention	26 A	Stiefel orthogonal bases	decoder-output calibration	static compressed cache	online residual and kernels
Swift-SVD	26 P	activation-aware spectral low rank	covariance aggregation + rank allocation	training-free weight/KV reduction	matrix objective; no sparse tensor tail
RAP	26 A	RoPE-pair column pruning	static	absorbed, no reconstruction	data-adaptive bulk/residual coding
STAR-KV	26 A	adaptive low rank + mixed precision	head/block thresholds	Triton path	independent reproduction; sparse innovations
KV-CoRE	26 A	rank diagnostic/benchmark	data/domain dependent	incremental SVD analysis	policy using diagnostics online
Quant. Dominates	26 A	matched-byte comparison	none	evaluation critique	test tensor+sparse direct methods
JoLT	26 A	partial Tucker + low-bit JL residual	joint rank/bit allocation	fast randomized fitting	direct attention; longer payback study
Tucker-KV	26 U	Tucker HOSVD/HOOI	refinement and budgeting	representation study	not archival; no novelty by Tucker alone
VarRate	26 A	variable-rate low rank	token/block rate control	training-free policy	independent reproduction and kernels

6 What the literature establishes—and what it does not

Question	Evidence-supported answer	Unresolved part
Is KV low rank?	Often, but unevenly across K/V, layers, heads, prompts, domain, and RoPE placement.	No universal rank; an online policy must detect non-compressible blocks.
Does smaller storage mean faster decode?	Only when fewer HBM bytes are read and reconstruction/update overhead is controlled.	Kernel crossover depends on rank, S, batch, head type, layout, and hardware.
Static or online?	Static/weight-side is cheaper; online bases better track distribution shift.	The best split between a fixed bulk and cheap online correction.
Frobenius or task-aware loss?	Attention/output-aware objectives better match model behavior.	A cheap causal estimator of future-query influence.
Matrix or tensor?	Multiway axes offer structure, but Tucker/CP/TT overhead can erase the gain.	Which modes are actually compressible and directly executable.
Low rank or quantization?	Quantization is a mandatory strong baseline and can dominate simple rank reduction at matched bytes.	Whether structured tensor+sparse methods create a new Pareto point at high dimension/long context.

Six defensible research gaps

1. No established method combines a CP-structured multiway bulk with a *RoPE-pair* group-sparse error chosen by predicted attention-output influence.
2. Existing residual hybrids largely minimize entrywise, quantization, or energy error; they do not cheaply optimize the risk of changing softmax routing under future queries.
3. Tensor papers focus primarily on representation/fitting; direct factor-domain attention and end-to-end kernel crossover remain underdeveloped.
4. Online low-rank work adapts bases or ranks, but rarely makes compression admission depend on the number of future cache reads needed to recover the fit cost.
5. K and V have different roles and spectra, yet many decompositions use symmetric modes, ranks, residual criteria, and precision.
6. Evaluation still over-rewards nominal compression. Physical bytes, metadata, fit cost, bytes/token, TPOT, energy, instruction preservation, and same-byte INT4 comparisons are often not reported together.

7 Recommended flagship: SAGE-CP-KV

Softmax-Aware Group-sparse Error-corrected CP KV cache

SAGE-CP-KV stores the stable bulk of each old KV block as CP factors whose modes correspond to token, KV head, RoPE frequency pair, and pair component. It stores only a small set of attention-critical residual *groups* aligned to complete RoPE pairs. A fused kernel computes logits and value aggregation directly from CP coefficients and sparse groups. A recent exact window protects local reasoning, and a payback controller uses native INT4 whenever tensor fitting would not recover its cost.

7.1 Why this is the best first paper

The idea has a clear Technique-Paper identity and a stronger novelty boundary than "runtime Tucker":

- It uses genuine multiway structure rather than replacing SVD with another dense factorization.
- It differs from GEAR: the sparse term corrects *attention influence* in RoPE-pair groups, not scalar quantization outliers.
- It differs from JoLT: the bulk is CP with direct attention algebra, while the residual is sparse and routing-aware rather than a dense low-bit JL residual.
- It differs from SALS/ShadowKV: sparsity represents approximation failures, not merely a selected token subset requiring reconstruction/offload.
- It is more feasible than fully online robust tensor decomposition because the CP dictionary can be calibrated offline and only token coefficients plus a tiny residual are computed online.

7.2 Representation

For an aged block of n tokens, reshape each layer cache as

$$\mathcal{K}, \mathcal{V} \in \mathbb{R}^{n \times H_{\mathrm{kv}} \times P \times 2}, \qquad P = d/2,$$

where the last two modes preserve complete RoPE pairs. Use separate CP ranks R_K, R_V:

$$\widehat{\mathcal{K}}_{s,h,p,u} = \sum_{r=1}^{R_K} A^K_{s,r} B^K_{h,r} C^K_{p,r} D^K_{u,r} + S^K_{s,h,p,u},$$

$$\widehat{\mathcal{V}}_{s,h,p,u} = \sum_{r=1}^{R_V} A^V_{s,r} B^V_{h,r} C^V_{p,r} D^V_{u,r} + S^V_{s,h,p,u}.$$

B, C, D are calibrated per layer or layer group. Only token coefficients $A_{s,:}$ and sparse residual groups grow with context. S is group sparse over (s, h, p) and always retains both components $u \in \{0, 1\}$, preserving RoPE-pair geometry and enabling regular memory access. K and V may use different ranks, precision, and residual budgets.

7.3 Softmax-aware fitting and sparse innovations

Plain CP-ALS minimizes tensor Frobenius error, which can spend rank on harmless energy while missing a small perturbation that changes attention routing. Calibrate CP factors using recorded queries $\mathcal{Q}_\ell$ and a mixed objective

$$\begin{aligned}\min_{\Theta,S} \quad & \mathbb{E}_{q\sim\mathcal{Q}_\ell}\left\|\mathrm{Attn}(q,\mathcal{K},\mathcal{V})-\mathrm{Attn}(q,\widehat{\mathcal{K}},\widehat{\mathcal{V}})\right\|_2^2 \\ & +\lambda_K\left\|\mathcal{K}-\widehat{\mathcal{K}}\right\|_F^2+\lambda_V\left\|\mathcal{V}-\widehat{\mathcal{V}}\right\|_F^2+\tau_K\left\|S^K\right\|_{2,0,\mathrm{pair}}+\tau_V\left\|S^V\right\|_{2,0,\mathrm{pair}}.\end{aligned}$$

For online residual admission, use a cheap upper-bound score rather than rerunning the full model. Let $e^K_{s,h,p,:}$ and $e^V_{s,h,p,:}$ be a candidate pair residual, $\bar{q}_{h,p}$ a high quantile of calibration-query magnitude, and $\bar{\alpha}_{s,h}$ predicted/observed attention mass:

$$I_{s,h,p}=c_K\,\bar{\alpha}_{s,h}\,\bar{q}_{h,p}\left\|e^K_{s,h,p,:}\right\|_2+c_V\,\bar{\alpha}_{s,h}\left\|e^V_{s,h,p,:}\right\|_2.$$

Store the highest-I groups under a byte budget. A conservative floor always protects system instructions, delimiters, exact numbers/entities identified by a lightweight policy, and the recent window. The paper should call I an *influence surrogate*, not a proof of semantic importance.

7.4 Reconstruction-free attention algebra

For query head h and its associated KV head, define

$$Z^K_{h,r}(q)=B^K_{h,r}\sum_{p,u}q_{h,p,u}C^K_{p,r}D^K_{u,r}.$$

Then every old-token logit is

$$\ell_{s,h}=\frac{1}{\sqrt{d}}\left(\sum_r A^K_{s,r}Z^K_{h,r}(q)+\sum_{(p,u)\in\Omega^K_{s,h}}q_{h,p,u}S^K_{s,h,p,u}\right).$$

This reads an $n\times R_K$ coefficient matrix plus compact factors, not an $n\times d$ key matrix. After online softmax produces $\alpha_{s,h}$, first reduce over tokens,

$$g^V_{h,r}=\sum_s\alpha_{s,h}A^V_{s,r},$$

then form the current output vector

$$o_{h,p,u}=\sum_r g^V_{h,r}B^V_{h,r}C^V_{p,r}D^V_{u,r}+\sum_s\alpha_{s,h}S^V_{s,h,p,u}.$$

Only the current d-dimensional output is formed; no dense old-V cache is reconstructed. Blockwise log-sum-exp merges exact-hot, CP, sparse, and fallback INT4 blocks.

7.5 Feasible lifecycle and computation cost

1. **Offline calibration:** learn B, C, D, ranks, query statistics, and optionally a small coefficient encoder. This absorbs expensive CP optimization before deployment.

2. **Native hot window:** append recent K/V in BF16, FP8, or INT4. Do not factorize per token and do not alter the just-produced layer output.

3. **Age-triggered coding:** when a block leaves the window, compute A with a precomputed regularized pseudoinverse or a one-pass encoder; compute residual pair scores and keep top groups. A bounded one-iteration refinement is an ablation, not a dependency.

4. **Payback admission:** let C_{fit} be measured coding time, Δt the measured time saved per future decode read, and N_{future} a conservative remaining decode estimate. Admit CP coding only if

$$N_{\mathrm{future}}\Delta t>C_{\mathrm{fit}}+C_{\mathrm{margin}}.$$

   Otherwise retain the block in the fastest available native/INT4 representation.

Intuitive advantage

CP factors capture the repeated, globally shared structure across heads and frequency pairs; sparse innovations preserve rare local exceptions such as names, numbers, delimiters, and instruction-bearing features. Pair grouping respects RoPE and maps to contiguous GPU work. Direct factor attention turns representation compression into fewer HBM bytes per token. The payback gate prevents the common failure where a beautiful decomposition makes inference slower.

7.6 Memory model and target operating region

Ignoring amortized fixed factors, dense BF16 storage per token/layer is $4H_{\mathrm{kv}}d$ bytes. SAGE-CP-KV stores approximately

$$2b_A(R_K + R_V) + m_K(2b_S + b_I) + m_V(2b_S + b_I)$$

bytes per token/layer, where m_K, m_V are retained RoPE-pair groups and b_I includes compressed indices. Fixed CP factors and rank tables must be included in measured allocation. The first target should be a conservative 4–8× physical reduction with near-full task quality, not a maximal nominal ratio.

7.7 Theory targets

For bounded quantities and approximation errors

$$\|q\|_2 \le Q, \qquad \max_s \|V_s\|_2 \le M_V, \qquad \epsilon_K = \max_s \left\|K_s - \widehat{K}_s\right\|_2, \qquad \epsilon_V = \max_s \left\|V_s - \widehat{V}_s\right\|_2,$$

softmax Lipschitzness gives a target bound of the form

$$\left\|\mathrm{Attn}(q, K, V) - \mathrm{Attn}(q, \widehat{K}, \widehat{V})\right\|_2 \le c_{\mathrm{sm}} \frac{Q\epsilon_K}{\sqrt{d}} M_V + \epsilon_V,$$

with constants and norm choices stated precisely in the paper. The stronger contribution is a group-selection result: under a calibrated query envelope, selecting the largest influence-bound groups minimizes the stated separable upper bound at a fixed group budget. Do not claim global optimality for CP rank or future-query prediction.

7.8 Experiments that can support publication

Claim	**Experiment**	**Pass criterion before writing the claim**
Multiway structure exists	CP rank/conditioning curves on recorded post-RoPE K and V by layer, head type, context, and domain	Stable rank gives $\ge 4\times$ physical reduction on substantial layers
Sparse tail protects routing	Compare no residual, magnitude scalar, random pair, energy pair, and influence pair at equal bytes	Influence pairs improve attention-output and downstream quality consistently
Direct execution is real	Explicit reconstruction vs factor kernel vs FlashAttention/INT4 microbenchmarks	No old dense allocation; positive latency region at realistic S, B
Online cost is repaid	Report fit time and future-read crossover per block	Controller declines unprofitable cases and amortized overhead stays small
Quality is robust	LongBench, RULER, GSM8K/MATH, PG-19 or WikiText; multi-instruction IFEval and prompt-leakage checks	Near-full quality at the selected operating point across two model families

Claim	Experiment	Pass criterion before writing the claim
New Pareto point	Same physical bytes against KIVI/INT4, Palu, Eigen, TALE, OjaKV, SALS, Lexico, GEAR, JoLT	Better quality or TPOT at equal memory on at least one important regime, without hiding regressions

Recommended models are one GQA family (Llama-3.1/3.2 or Qwen2.5/3) and one MHA or different-GQA family (Mistral/Llama-2), subject to license and hardware. Sweep context 2K–128K where feasible, batch size, hot-window length, R_K/R_V, sparse budget, precision, and pre/post-RoPE coding. Report TTFT, TPOT, tokens/s, peak allocation, actual cache bytes, metadata, bytes read/token, fit time, energy/token, and accuracy. Use bootstrap confidence intervals or repeated seeds for task metrics.

Fatal-flaw gates

- If CP rank is high or unstable on values, pivot to CP keys + quantized values; do not force a symmetric method.
- If factor kernels do not beat INT4 at long context, publish only if the quality-at-memory gain is large and honestly position it as a capacity method.
- If influence selection does not beat pair magnitude at equal bytes, the claimed novelty is not supported.
- If online coefficient/residual coding cannot repay itself, move coding to prefill/offline weight-side generation or abandon runtime fitting.
- If gains disappear on a second model/domain, narrow the claim to a diagnosed compressibility regime rather than claiming universal KV structure.

8 Shared mathematical foundation of the proposed portfolio

All four ideas use the same high-dimensional modeling principle but apply it to different objects and system boundaries. For a D-way tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times \cdots \times n_D}$, we seek

$$\mathcal{X} = \mathcal{L} + \mathcal{S} + \mathcal{E}, \qquad \text{structured rank}(\mathcal{L}) \text{ small}, \quad \|\mathcal{S}\|_{2,0,\mathcal{G}} \le k, \quad \|\mathcal{E}\| \le \varepsilon,$$

where $\mathcal{L}$ captures recurring global structure, $\mathcal{S}$ contains a small number of exceptional groups from a hardware-aligned partition $\mathcal{G}$, and $\mathcal{E}$ is the unrepresented tail. This is a modeling hypothesis to be tested per layer, prompt, and modality, not assumed universally.

8.1 Three tensor representations under consideration

- **CP:**

$$\mathcal{L}_{i_1,\ldots,i_D} = \sum_{r=1}^{R} \prod_{d=1}^{D} U^{(d)}_{i_d,r}, \qquad \text{Bytes}(\mathcal{L}) = b \sum_{d=1}^{D} n_d R.$$

  CP has the smallest and most direct factor algebra, which is why it is the first choice for SAGE-CP-KV. Its main difficulty is numerical conditioning and the possible nonexistence of a stable best rank-R approximation.

- **Tucker:**

$$\mathcal{L} = \mathcal{G} \times_1 U^{(1)} \times_2 \cdots \times_D U^{(D)}, \qquad \text{Bytes}(\mathcal{L}) = b \left(\prod_d r_d + \sum_d n_d r_d \right).$$

  Tucker provides separate ranks per semantic mode and is often easier to fit, but the core $\mathcal{G}$ can grow exponentially with the number of jointly compressed modes. JoLT already occupies much of the direct Tucker novelty space.

- **Tensor Train:**

$$\mathcal{L}_{i_1,\ldots,i_D} = G^{(1)}[:, i_1, :] G^{(2)}[:, i_2, :] \cdots G^{(D)}[:, i_D, :], \qquad \text{Bytes}(\mathcal{L}) = b \sum_{d=1}^{D} r_{d-1} n_d r_d,$$

  with $r_0 = r_D = 1$. TT is attractive for causal/multiscale token tensorization and for LoSpar-CI/CATT-KV, but ranks can grow under appends, permutations, or poorly chosen mode order.

The sparse term is deliberately *group* sparse rather than arbitrarily element sparse:

$$\|\mathcal{S}\|_{2,0,\mathcal{G}} = \sum_{g\in\mathcal{G}} \mathbf{1}\{\|\mathcal{S}_g\|_2 > 0\}.$$

Candidate groups include RoPE coordinate pairs, feature tiles, tokens, attention heads, spatial patches, or hyperspectral bands. Grouping slightly reduces the ideal compression ratio but makes the representation semantically meaningful and executable by regular GPU/edge kernels.

Why low rank and sparsity are complementary

Low rank explains the *shared bulk*: correlations repeated across tokens, heads, frequency pairs, spatial patches, or devices. Sparsity explains *rare innovations*: prompt-specific entities, numbers, instructions, pathological regions, unusual spectral signatures, and distribution-shift errors that a fixed subspace misses. Low rank alone can blur rare information; sparsity alone repeatedly stores common structure. Their combination is useful only when the sparse support remains small and the factor/residual kernels are cheaper than dense movement.

9 LoSpar-CI: progressive tensor factors plus sparse refinement

Proposed cross-domain technique

LoSpar-CI proposes to transmit the intrinsic structure of a high-dimensional intermediate tensor across collaborating devices: send tensor factors for a coarse low-rank component first, then request sparse innovation groups only while the downstream decision is not certifiably stable.

9.1 Intended setting and research gap

The strongest LoSpar-CI setting is not batch-1 LLM decode activation transfer; a single token crossing a pipeline boundary is only one hidden vector and its network cost may be dominated by round-trip/framework latency. The natural setting is an actually high-dimensional boundary tensor: long-prompt prefill activations, hyperspectral cubes, dynamic MRI/CT, long video, 3D volumes, or whole-slide-image features. Robust PCA and tensor decompositions provide the mathematical foundation for low-rank-plus-sparse structure [1, 2, 3]; HexGen, Jupiter, and the FSU edge-pipeline poster provide the collaborative-inference setting [6, 7, 8].

9.2 Representation and communication complexity

At pipeline cut ℓ, let

$$\mathcal{A}_\ell = \mathcal{L}_\ell + \mathcal{S}_\ell + \mathcal{E}_\ell, \qquad \|\mathcal{E}_\ell\| \le \varepsilon_0.$$

For a TT implementation, device d_p transmits cores $\{G_\ell^{(j)}\}_{j=1}^{D}$ at cost

$$C_L = b_L \sum_{j=1}^{D} r_{j-1} n_j r_j.$$

Let k of N_G fixed-size groups be retained in $\mathcal{S}_\ell$. A direct support/value code costs approximately

$$C_S(k) = k\, b_{\text{group}} + \left\lceil \log_2 \binom{N_G}{k} \right\rceil \lesssim k\, b_{\text{group}} + k \log_2\left(\frac{eN_G}{k}\right).$$

The total is $C_L + C_S(k)$ rather than $b\prod_j n_j$, but only after factor headers, indices, padding, quantization scales, and the cost of fitting $\mathcal{L}_\ell$ are counted.

9.3 Progressive protocol and a correct decision certificate

Let $\widehat{\mathcal{A}}_\ell^{(j)} = \widehat{\mathcal{L}}_\ell + \mathcal{S}_\ell^{(j)}$ after j sparse groups, and assume a computable residual bound

$$\left\|\mathcal{A}_\ell - \widehat{\mathcal{A}}_\ell^{(j)}\right\| \le \eta_j.$$

Let F_ℓ map the cut activation to final logits and satisfy locally

$$\|F_\ell(A) - F_\ell(A')\|_\infty \le \kappa_\ell \|A - A'\|.$$

If the approximate top-two logit margin is $\widehat{\mu}_j = \widehat{y}_{(1)} - \widehat{y}_{(2)}$, then

$$\widehat{\mu}_j > 2\kappa_\ell \eta_j \quad \Longrightarrow \quad \arg\max F_\ell(\mathcal{A}_\ell) = \arg\max F_\ell(\widehat{\mathcal{A}}_\ell^{(j)}).$$

The factor two protects both the winning and runner-up logits. The sender transmits low-rank factors first; the receiver stops if the inequality holds, otherwise it requests the next highest-priority residual group. For generation, a strict global certificate is probably too costly per token; there it should become an offline-calibrated error-budget allocator rather than a repeated downstream recomputation.

9.4 What would constitute the paper

1. A modality-specific demonstration that selected boundary tensors have stable TT/Tucker rank plus a concentrated group-sparse tail.
2. A progressive factor/residual wire protocol with measured encoding, metadata, and recomputation costs.
3. A certified or statistically calibrated early-stop rule that determines how many residual groups each sample requires.
4. An end-to-end multi-device evaluation showing that bytes and latency scale with intrinsic rank/sparsity rather than ambient tensor dimension.

Expected advantages

Communication can fall dramatically when ambient dimension grows while TT/Tucker ranks and innovation count remain small. Progressive transmission adapts quality per example instead of committing to one lossy compression ratio. The sparse tail protects rare, high-value information that low rank tends to smooth away. A valid margin certificate provides a much stronger quality story than an average reconstruction metric.

Possible limitations

The structural assumption can fail at some cuts, layers, or modalities; TT ranks and sparse support may be too large. Tensor fitting can exceed the communication time saved. Lipschitz bounds may be so loose that almost no sample stops early, and repeated refinement adds round trips. For LLM batch-1 decoding, boundary-message size is often not the bottleneck. The first LoSpar-CI paper should therefore choose one demonstrably compressible high-dimensional modality or prefill, profile decomposition payback, and keep a dense fallback.

10 Shufflerless-CI: stationary latent–residual collaborative decoding

Proposed systems co-design

Shufflerless-CI proposes to make KV state stationary on the device that owns each layer. A fixed RoPE-safe low-rank projection creates the bulk cache, a sparse prompt-specific residual corrects aged blocks, and a heterogeneous placement planner admits only requests whose weights, cache, and workspace fit without CPU–GPU or inter-device KV movement.

10.1 Born-compressed bulk and online sparse innovations

For layer ℓ, factor projection weights offline:

$$W_\ell^K \approx U_\ell^K (P_\ell^K)^\top, \qquad W_\ell^V \approx U_\ell^V (P_\ell^V)^\top.$$

For hidden state x_s, cache only

$$z_s^K = x_s U_\ell^K, \qquad z_s^V = x_s U_\ell^V,$$

plus a small exact hot window. When a full block B ages, compute once

$$E_B^K = K_B - \widehat{K}_B(z_B^K), \qquad E_B^V = V_B - \widehat{V}_B(z_B^V),$$

retain hardware-aligned top-$k_{\ell h}$ innovation groups, and discard the dense block. This avoids a per-token or accumulated-cache SVD. RoPE requires either complete coordinate-pair projection, as motivated by RAP, or a small exact positional branch plus semantic latent, as in MLA/MHA2MLA [31, 25, 15].

10.2 Direct attention and exact hot/cold merge

For a compatible factorization, the old-block score and value contribution have the form

$$s_{tB} \approx (q_t P_\ell^K)(z_B^K)^\top + q_t (E_B^K)^\top,$$

$$\alpha_B V_B \approx \big(\alpha_B z_B^V\big)(P_\ell^V)^\top + \alpha_B E_B^V.$$

The dense old K_B, V_B are never allocated. Exact-hot and approximate-old segments produce block statistics

$$m_j = \max_s s_{js}, \quad \ell_j = \sum_s e^{s_{js}-m_j}, \quad o_j = \sum_s e^{s_{js}-m_j} v_{js}.$$

Following online softmax/FlashAttention [4, 5], their exact merge is

$$m = \max_j m_j, \qquad \ell = \sum_j e^{m_j - m}\ell_j, \qquad o = \sum_j e^{m_j-m} o_j, \qquad \text{output} = o/\ell.$$

This merge adds no approximation; all error is isolated to the compressed old-block representation.

10.3 Joint placement and deterministic no-shuffle admission

For devices $i = 1, \dots, P$, the planner chooses the contiguous layer partition π, ranks r_ℓ, sparse budgets k_ℓ, bit widths b_ℓ, and hot windows W_ℓ:

$$\min_{\pi,r,k,b,W} \max_i T_i(\pi, r, k, b, W)$$

subject to

$$M_i^{\text{weights}}(\pi) + M_i^{\text{hot}}(W) + M_i^{\text{old}}(r,k,b) + M_i^{\text{workspace}} \le M_i,$$

$$\sum_{\ell,h} \mathcal{E}_{\ell h}(r_{\ell h}, k_{\ell h}, b_{\ell h}) \le \epsilon.$$

For an admitted maximum sequence length, these inequalities make "no shuffle" a deterministic capacity property rather than a hope. The implementation must verify it with explicit-copy, page-migration, page-fault, and synchronization traces, especially on unified-memory Jetsons.

Expected advantages

The method directly targets the memory wall and tail latency: each layer reads a compact local cache, never waits for remote KV, and never allocates a dense reconstruction buffer. Fixed weight-side factors avoid runtime decomposition, while sparse innovations provide prompt-specific adaptation. Joint rank/placement allocation can rebalance a heterogeneous pipeline and admit longer contexts than the fixed dense partition.

Possible limitations

The decomposition primitives are close to Palu, Eigen Attention, RAP, OjaKV, and other hybrids; novelty must come from the stationary representation–placement co-design and a decisive fixed- versus-joint-placement ablation. Sparse indexing or RoPE processing may erase kernel gains. A fixed subspace may fail under domain shift. Admission control can reject long or uncertain requests rather than serving them. Finally, the project requires a multi-device implementation, so it is less feasible as the first paper than SAGE-CP-KV.

11 SparsePipe: sparsity as a schedulable distributed resource

Proposed adaptive systems layer

SparsePipe proposes to choose block sparsity, compression primitive, wire format, and pipeline allocation jointly from live compute, bandwidth, memory, and quality state. It treats sparsity as a runtime resource rather than a fixed model hyperparameter and always retains a dense/exact fallback.

11.1 Mask prediction and joint control problem

At boundary p and control interval t, a lightweight predictor produces block scores $a_{p,t} = g_{\theta_p}(x_{p,t})$ and the controller selects

$$m_{p,t,g} = \mathbf{1}\{a_{p,t,g} \geq \tau_{p,t}\}, \qquad g = 1, \ldots, N_p^{\text{blk}}.$$

The action σ_t contains mask thresholds, token/KV budget, representation type, bit width, wire format, and a *coarse-timescale* partition choice. The optimization target is

$$\min_{\sigma_t} \; \alpha\,\text{TTFT}(\sigma_t) + \beta\,\text{TPOT}(\sigma_t) + \gamma\,\text{Energy}(\sigma_t) + \delta\,\text{Memory}(\sigma_t) + \zeta \max_i \text{Latency}_i(\sigma_t)$$

subject to per-device memory and

$$\text{QualityRisk}(\sigma_t) \leq \rho.$$

The partition should not move every token: moving weights or KV would contradict the latency goal. Fast control changes masks/formats; slow control changes partitions only between requests or at rare safe synchronization points.

11.2 A wire-format crossover that counts metadata

If each block contains G values of b_v bytes, N_{blk} blocks exist, and n_a are active, then

$$C_{\text{dense}} = N_{\text{blk}} G b_v, \qquad C_{\text{sparse}} = n_a G b_v + C_{\text{index}}(n_a, N_{\text{blk}}).$$

Sparse transmission is selected only when $C_{\text{sparse}} < C_{\text{dense}}$ and the profiled sparse kernel plus packing time is faster. With a bitmap, $C_{\text{index}} = N_{\text{blk}}/8$ bytes; with explicit indices it is approximately $n_a \lceil \log_2 N_{\text{blk}} \rceil / 8$. This simple crossover prevents the controller from choosing "sparsity" that increases total traffic.

11.3 Composed accuracy budget

Suppose an approximation error Δ_p is injected at boundary p and the downstream suffix $F_{p+1:P}$ has local Lipschitz constant $\kappa_{p+1:P}$. For multiple boundaries,

$$\|\Delta_{\text{out}}\| \leq \sum_{p=1}^{P-1} \kappa_{p+1:P} \, \|\Delta_p\| \, .$$

SparsePipe allocates a conservative error budget ρ_p so that $\sum_p \kappa_{p+1:P} \rho_p \leq \rho$. Because worst-case constants can be loose, the deployable version should combine calibrated empirical risk with exact fallback and report both the formal bound and its coverage/tightness.

11.4 How tensor and low-rank methods enter SparsePipe

SparsePipe is a controller, not a replacement for SAGE-CP-KV or LoSpar-CI. Its action set can select among dense, INT4, CP-only, CP+sparse, TT/Tucker factors, token eviction, or exact protected storage. For high-dimensional prefill boundaries it can choose LoSpar-CI's factor/residual wire format; for local KV it can choose SAGE-CP-KV; for the multi-device steady-state guarantee it can respect Shufflerless-CI's stationary placement. This composability is the most defensible positioning.

Expected advantages

The method can respond to bandwidth drops, thermal throttling, queue changes, and heterogeneous devices instead of using one compression ratio everywhere. Block granularity and format switching make sparsity hardware-aware. Joint control can spend quality budget where it most reduces the pipeline bottleneck, and exact fallback provides a safety floor. The controller can reuse advances in tensor, quantization, eviction, or sparse coding as interchangeable actions.

Possible limitations

The scope is broad: token, neuron, attention, KV, wire format, and placement decisions can make credit assignment and evaluation unwieldy. Contextual sparsity predictors can fail under shift, and conservative quality bounds may force dense execution. At batch-1 decode, activation bytes may be too small for sparsification to overcome network round-trip latency. Sparse kernels, packing, and indices often cost more than idealized FLOP models suggest. Per-token repartitioning is infeasible. SparsePipe should therefore begin with one or two actions, a profiled model- predictive/bandit controller, and a static-sparsity plus fixed-partition ablation.

12 Portfolio comparison and recommended sequencing

Idea	Compressed object	Tensor / low-rank role	Sparse role	Primary advantage	Main limitation
SAGE-CP-KV	Old KV on one accelerator	CP across token/head/RoPE pair/component	Influence-selected RoPE-pair corrections	Direct factor attention; strongest first-paper balance	CP conditioning and kernel crossover
LoSpar-CI	High-dimensional boundary activation	TT/Tucker/CP factors transmitted first	Progressive exceptional groups	Intrinsic-complexity communication with certificate	Loose certificate; refinement/fitting overhead
Shufflerless-CI	Per-layer KV across an edge pipeline	Fixed born-compressed latent	Aged-block prompt innovations	Deterministic stationary cache and lower tail latency	Multi-device complexity; novelty is systems coupling
SparsePipe	Multiple runtime objects and formats	Selects tensor/latent codecs as actions	Schedules block masks and exact fallback	Adapts to live heterogeneity and bandwidth	Very broad scope; predictor/kernel overhead
FD-KV	Streaming KV basis on one accelerator	Deterministic Frequent-Directions sketch of the key stream, query-whitened	Optional attention-weighted V metric	Provable online basis at fixed update cost; beats heuristic Oja	Tracks only the dominant subspace
SC-KV	Aged KV blocks on one accelerator	Union of low-rank subspaces (block-term) per token cluster	Per-cluster residual groups	Captures multi-domain heterogeneity one subspace misses	Cluster metadata; hard assignment needed for direct execution

Recommended sequence

Phase 1: test SAGE-CP-KV on recorded single-device KV and build the direct CP+sparse kernel. **Phase 2:** if the kernel yields a real Pareto improvement, integrate it into Shufflerless-CI and jointly optimize representation with the poster's layer partition. **Parallel/alternative branch:** test LoSpar-CI on one genuinely high-dimensional modality where $L + S + E$ is well supported. **Phase 3:** add SparsePipe only after two or more representation choices have measured operating regions worth scheduling.

13 Companion ideas, ranked

13.1 Idea 2: Payback-KV—a heterogeneous representation controller

Choose per block among native FP16/FP8, INT4, CP-only, CP+sparse, and exact protected storage using measured fit cost, predicted remaining output length, rank diagnostics, and kernel latency tables. The contribution is a decision policy over *realized latency and quality risk*, not another compression format. It is highly feasible and systems-friendly, but its novelty is moderate unless the controller has a principled regret/risk formulation and demonstrates broad hardware transfer. Best fit: MLSys/ASPLOS/EuroSys; use it as a controller contribution around SAGE-CP-KV rather than a standalone NeurIPS method initially.

13.2 Idea 3: CATT-KV—causal Tensor-Train with sparse innovations

Tensorize old token positions into dyadic causal digits and maintain SVD-free TT cores. Derive RoPE-equivariant core contractions and store attention-critical sparse innovations. This can model multiscale sequence structure better than CP and has the highest algorithmic upside. It is also the riskiest: TT ranks may grow, appends and block merges are complex, and direct softmax still needs all token logits unless combined with hierarchical screening. Pursue after the CP pilot shows that tensor-domain execution has a positive crossover. Best fit if successful: NeurIPS/ICML for the method plus MLSys/ASPLOS for a kernel-heavy version.

13.3 Idea 4: protected-instruction tensor cache

Use a learned or certified policy to route system instructions, delimiters, and low-frequency semantic anchors to exact/sparse storage while factorizing ordinary context. The gap is motivated by instruction-loss and prompt-leakage evidence [39]. It becomes publishable only if the protection rule generalizes beyond hand-written token classes and improves safety at matched bytes; otherwise it is an evaluation ablation, not a full paper.

13.4 Idea 5: distribution-shift-triggered tensor adaptation

Keep offline CP factors but monitor a cheap residual/influence statistic. Only when it exceeds a threshold, update a small factor subset or open new rank components; retire them when no longer useful. This hybrid addresses the static-basis weakness identified by OjaKV without continuous adaptation. The hard novelty question is whether the trigger and update outperform Oja's matrix subspace at matched update cost. Treat this as a second paper or SAGE-CP-KV extension.

13.5 Ideas 6–7: open-corner methods from the tensor-decomposition literature

Two directions survive a strict novelty–feasibility–publishability filter against the surveyed landscape; both target the open problems of §4.1 (static-basis distribution shift, Frobenius-versus-output-aware objective, K/V asymmetry) and the six gaps of §6 using decomposition machinery not yet applied to KV caches. They are added to the comparison table (§12) and the evaluation (§14). Weaker corners are merged into them or explicitly not pursued (end of this subsection).

Idea 6: FD-KV—a provable, output-aware streaming basis

Replace the static subspace of §4.1 (Eigen, MatryoshkaKV) and the heuristic online PCA of OjaKV with a **Frequent-Directions** sketch [43, 44]: per layer and KV head, maintain an $\ell \times d$ sketch B of the streaming key rows whose top-r right singular vectors form the compression basis. FD is deterministic and satisfies $0 \preceq K^\top K - B^\top B$ and $\left\|K^\top K - B^\top B\right\|_2 \le \|K - K_r\|_F^2 / (\ell - r)$—a worst-case bound on the exact Gram/feature subspace attention projects onto, at fixed $O(d\ell)$ update cost, a guarantee Oja's stochastic-gradient PCA lacks. *Two refinements make the objective output-aware and K/V-asymmetric*, absorbing what would otherwise be a separate, weaker idea: run FD on query-whitened key rows qK so the bound becomes a score-fidelity bound (§6 gap 2), and sketch V in an attention-mass-weighted metric because $o = \sum_s \alpha_s V_s$ weights value error by attention (§6 gap 5). **Closest prior:** OjaKV (online PCA, no worst-case error), STAR-KV (soft-threshold ranks), and KQ-SVD (query-aware K, but offline and matrix-only); the streaming-attention discrepancy sketch [50] is adjacent but targets the attention product, not a cached basis. **Novelty:** a deterministic, guaranteed, query-whitened streaming KV basis with asymmetric K/V metrics at matched update cost—the comparison Idea 5 calls unresolved. **Feasibility:** very high (a few lines; no offline calibration required). **Falsification:** FD-basis versus Oja versus static SVD under domain shift, at equal bytes and equal update cost. **Risk:** it tracks only the dominant subspace and degrades—rather than fails—when KV is genuinely high rank.

Idea 7: SC-KV—subspace-clustered KV via block-term decomposition

A single global low-rank subspace blurs the heterogeneity that makes "no universal rank" true (§6 Q&A). Model each aged block as a **union of low-rank subspaces**—a block-term decomposition [45] with rank-$(L, L, 1)$ terms—by clustering old tokens into a few groups and giving each group its own small subspace. With a *hard* token-to-cluster assignment this is directly executable: per-cluster scores come from per-cluster factors and merge by the same online-softmax argument used elsewhere, with no dense reconstruction. It addresses distribution shift *representationally* (complementary to FD's temporal adaptation) and the "which modes are compressible" question (gap 3, §4.1). **Closest prior:** every §4.1 method and xKV/OjaKV fit *one* subspace per head/layer; no surveyed KV method uses a block-term or union-of-subspaces representation. **Feasibility:** moderate—k-means on keys plus per-cluster SVD is easy offline, but online cluster assignment, metadata, and block kernels need care. **Falsification:** on a long multi-section prompt, do k subspaces of rank r/k beat one rank-r subspace at equal bytes on attention-output error? **Risk:** cluster-boundary metadata and assignment cost; keep assignments hard to preserve direct execution.

Merged and not-pursued corners

Merged. A fast randomized coefficient solve (TensorSketch [46]) is not a representation but a fit-cost *enabler*; fold it into the SAGE-CP-KV / SC-KV online coding step to help the payback gate clear (§6 gap 4) rather than ranking it. The query-whitened K and attention-weighted V metrics are merged into FD-KV above, since on their own they are close to KQ-SVD / Swift-SVD. **Not pursued as standalone.** A feature-axis interpolative

decomposition (keep some exact channels, interpolate the rest) is too close to CurDKV (row CUR) and RAP (RoPE-pair pruning) to be a distinct paper. A cross-layer variant that shares the growing token-loadings factor is subsumed by xKV [48] and MiniCache [49]. A **t-SVD** along the RoPE-frequency mode [47] is a genuinely novel but speculative question—does RoPE's Fourier structure induce spectral low rank?—worth a one-day spectrum check, not a plan.

Which §6 gaps the two surviving ideas close

FD-KV: static-basis shift with a bound (gap 1), amortization (gap 4), and—via the merged query-whitened / attention-weighted metrics—the output-aware objective and K/V asymmetry (gaps 2, 5). SC-KV: heterogeneity / "no universal rank" as a representation (gap 1, by a different mechanism) and directly executable multiway structure (gap 3). Neither removes the §6 gate that a method must beat INT4 at matched physical bytes; FD-KV is the likeliest to, because it changes the *amortization*, not only the ratio.

14 Portfolio evaluation

Scores below follow the five-dimension order Higher, Faster, Stronger, Cheaper, and Broader.

Idea	H	F	S	C	B	Verdict
SAGE-CP-KV	4/5	4/5	4/5	4/5	4/5	**Pursue first.** Best balance of novelty, single-device feasibility, tensor identity, and measurable system benefit.
Payback-KV	3/5	5/5	4/5	5/5	5/5	Strong systems module; alone may look like policy engineering.
CATT-KV	5/5	3/5	4/5	3/5	4/5	High-risk/high-reward. Gate on TT rank and kernel microbenchmarks before model integration.
Protected cache	4/5	3/5	5/5	4/5	3/5	Valuable safety layer; needs a generalizable mechanism, not token heuristics.
Triggered adaptation	4/5	3/5	4/5	3/5	4/5	Promising follow-up; closest-prior burden against OjaKV/STAR-KV is high.
LoSpar-CI	4/5	3/5	4/5	4/5	5/5	Strong high-dimensional collaborative-inference direction if measured tensors exhibit a stable low-rank bulk and a genuinely sparse, useful tail.
Shufflerless-CI	3/5	4/5	4/5	5/5	4/5	The most direct systems attack on the memory wall; publishability depends on a fused stationary-format kernel and end-to-end tail-latency wins.
SparsePipe	3/5	4/5	4/5	4/5	5/5	Broadest systems vision, but also the largest scope. Pursue after one codec and its cost model are empirically stable.
FD-KV	4/5	5/5	5/5	5/5	4/5	**Strongest feasible add.** Deterministic query-whitened streaming basis with a guarantee Oja lacks; gate on beating OjaKV and INT4 at matched update cost and bytes.
SC-KV	5/5	3/5	4/5	3/5	4/5	High-reward representational idea for heterogeneity; gate on whether clustered subspaces beat one subspace at equal bytes with direct execution.

Paradigm test

The project is incremental if framed as "CP instead of SVD." It becomes a potentially strong paper when the unit of design changes from a decomposition algorithm to an *attention-executable memory representation*: factor bulk, structured exceptions, positional geometry, online admission, and kernel are optimized together. The most defensible headline is therefore not higher compression alone, but **routing-stable tensor compression that reduces real bytes read per decode token without dense reconstruction**.

15 A staged research plan

1. **Weeks 1–2: falsify cheaply.** Record K/V/query tensors from two models and three domains. Plot mode unfoldings, CP error/conditioning, attention-output error, and INT4 baselines. Stop if no low-rank operating

region exists.

2. **Weeks 3–4: validate the novel residual.** Implement offline CP plus equal-byte residual selectors. Show that influence-pair selection improves routing/output relative to magnitude, scalar sparsity, and random groups.

3. **Weeks 5–7: exact algebra prototype.** Compare direct CP equations with explicit reconstruction numerically. Implement blockwise online softmax and sparse corrections in PyTorch, then profile.

4. **Weeks 8–11: fused GPU kernel.** Build Triton/CUDA kernels only after the profiler shows a plausible bandwidth crossover. Include metadata and ragged-sparsity costs.

5. **Weeks 12–15: model integration and robust evaluation.** Add exact hot window, payback admission, INT4 fallback, two model families, long-context/reasoning/instruction tests, and confidence intervals.

6. **Weeks 16–18: paper.** Lead with the diagnosed gap and measured crossover, not the acronym. Release kernels, memory accounting scripts, recorded-tensor pilot protocol, and negative regions.

15.1 How the broader portfolio should branch

The schedule above is deliberately centered on SAGE-CP-KV because it supplies the representation and kernel evidence needed by the other projects. After the Week 5–7 algebra prototype, the portfolio should branch by observed bottleneck rather than by acronym:

- If multiway low-rank structure is stable but the sparse residual carries essential rare features, generalize the codec and certificate into **LoSpar-CI**.
- If device memory traffic and KV migration dominate even after compression, freeze the representation into a stationary latent format and pursue **Shufflerless-CI**.
- If sparsity and bandwidth vary substantially across samples, layers, or links, expose codec choice as a control variable and pursue **SparsePipe**.

This ordering prevents SparsePipe from becoming a scheduler in search of a proven primitive and prevents Shufflerless-CI from claiming a no-movement advantage for a representation whose fused execution has not yet crossed the dense baseline.

16 Venue positioning and claim discipline

NeurIPS/ICML/ICLR: emphasize the attention-influence objective, group-selection theory, CP representation, and generalization across models/domains; system speed is supporting evidence. **MLSys/ASPLOS/EuroSys:** emphasize direct kernels, payback admission, physical bytes, roofline/crossover analysis, and workload-aware results; the representation is the enabling mechanism. **ACL/EMNLP/TACL:** emphasize long-context quality, reasoning, instruction preservation, and model behavior, with strong efficiency evidence.

No venue can be guaranteed. Avoid "first tensor KV compression," "near-lossless" without uncertainty, or "real-time" without measured fit amortization. A safe novelty claim, if supported, is:

> We introduce a CP-structured KV representation with RoPE-pair group-sparse innovations selected by a causal attention-influence surrogate, and a fused attention algorithm that consumes both terms without reconstructing old keys or values. A payback controller selects the representation per block using measured coding and decode costs.

17 Conclusion

The literature supports low-dimensional KV structure, but not a universal rank or a free runtime win. The most competitive research must survive three tests simultaneously: it must preserve softmax routing and downstream behavior, beat strong low-bit baselines at matched physical bytes, and reduce end-to-end bytes/time after fitting and metadata. Our recommended first direction is SAGE-CP-KV: it turns our high-dimensional sparsity and tensor-decomposition interests into a precise gap—shared multiway bulk plus rare structured innovations, executed as a memory system rather than reconstructed as a dense approximation. LoSpar-CI is the principled generalization to collaborative inference, Shufflerless-CI is the stationary-format systems realization for the memory wall, and SparsePipe is the eventual controller that makes representation sparsity an explicit schedulable resource. Together they form a coherent sequence, not four independent claims.

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